

DATA ANALYTICS PROJECT

Sales Data Analysis and Reporting for Retail Chain

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Course: Data Analytics

Kaggle – Retail Transaction Data
125,000 Rows | 2011–2015

Tech Stack -Python · SQL (SQLite) · Microsoft Excel

Libraries - Pandas · Matplotlib · Seaborn · OpenPyXL Data Source

1. PROJECT OVERVIEW

1.1 Project Aim

The aim of this project is to use Python, SQL, and Excel to analyze sales data and generate meaningful reports for a retail chain. The project covers the complete data analytics pipeline — from raw data ingestion through a structured database, rigorous cleaning, insightful analysis, and finally automated Excel reporting.

1.2 Objectives

- Set up a structured SQLite database using SQL DDL commands to store transaction data
 - Clean and validate 125,000 transaction records using Python pandas and SQL queries
 - Perform exploratory and advanced data analysis using pandas, matplotlib, and seaborn
 - Generate 5 automated chart visualisations showing sales trends and patterns
 - Produce a formatted 6-sheet Excel dashboard with KPIs, tables, and embedded charts
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1.3 Tech Stack

Component	Details
Language	Python 3.14
Database	SQLite (sqlite3 module)
Data Manipulation	Pandas 3.0, NumPy 2.4
Visualization	Matplotlib 3.10, Seaborn 0.13
Excel Generation	OpenPyXL 3.1
SQL	DDL + DML via sqlite3
IDE	Visual Studio Code
OS	Windows 11

1.4 Data Source

Source: Kaggle – Retail Transaction Data

Column	Description
customer_id	Unique identifier for each customer
trans_date	Transaction date (DD-Mon-YY)
tran_amount	Transaction amount (£)
ItemCode	15 product types (generated)
Country	8 countries (generated)

2. PHASE 1 – DATA COLLECTION & DATABASE SETUP

2.1 Data Collection

The dataset was downloaded from Kaggle as a CSV file containing **125,000 retail transactions** spanning **May 2011 to March 2015**.

2.2 Database Schema Design

A **star-schema SQLite database** was created with:

- 1 Fact Table
- 3 Dimension Tables

Table	Description
fact_sales	Core transaction data
dim_item	Product master (15 items)
dim_country	Country lookup (8 countries)
dim_date	Date dimension (1,401 records)

2.3 SQL Schema (Key Table)

```
CREATE TABLE fact_sales (  
    transaction_id    TEXT PRIMARY KEY,  
    transaction_time  TEXT NOT NULL,  
    date_id           TEXT NOT NULL,  
    item_code         TEXT NOT NULL,  
    country_id        INTEGER NOT NULL,  
    number_of_items   INTEGER NOT NULL,  
    cost_per_item     REAL NOT NULL,  
    total_sales_value REAL GENERATED ALWAYS AS  
                        (number_of_items * cost_per_item) STORED  
);
```

2.4 Phase 1 Results

Metric	Result
Total Rows	125,000
Products	15
Countries	8
Date Records	1,401
Execution Time	~10 seconds

3. PHASE 2 – DATA CLEANING & PREPARATION

3.1 Data Cleaning Steps

- No duplicate records found
- No missing values
- Outlier detection using IQR method
- Outliers capped (not removed)
- Data types validated using pandas

3.2 Outlier Detection Code

```
def flag_outliers(series):  
    Q1, Q3 = series.quantile(0.25), series.quantile(0.75)  
    IQR = Q3 - Q1  
    lower, upper = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR  
    return (series < lower) | (series > upper)
```

3.3 New Features Added

Field	Description
total_sales_value	items × cost
hour_of_day	extracted from timestamp
week_number	ISO week
date_id	formatted date

3.4 Summary Statistics

Metric	Value
Mean	£64.99
Std Dev	£22.86
Min	£10
Max	£105

4. PHASE 3 – DATA ANALYSIS

4.1 SQL Query – Quarterly Revenue

```
SELECT d.year, d.quarter,  
       COUNT(*) AS transactions,  
       SUM(f.total_sales_value) AS revenue,  
       AVG(f.total_sales_value) AS avg_order_value  
FROM fact_sales f  
JOIN dim_date d ON f.date_id = d.date_id  
GROUP BY d.year, d.quarter;
```

4.2 Key Results

- Total Revenue: **£8,123,989**
 - Avg Order Value: **£64.99**
 - Top Product: **USB Keyboard**
 - Top Country: **United Kingdom (~35%)**
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4.3 Charts Generated

1. Monthly Revenue Trend
 2. Top Products
 3. Revenue by Country
 4. Heatmap (Month × Day)
 5. MoM Growth
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4.4 Insights

- Revenue is highly stable
 - Slight Q4 seasonal increase
 - No extreme fluctuations
 - Strong dominance from UK market
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5. PHASE 4 – EXCEL DASHBOARD

5.1 Workbook Structure

Sheet	Description
KPI Dashboard	Summary metrics + charts
Monthly Sales	Monthly breakdown
Top Products	Product ranking
Country Analysis	Revenue by country
Quarterly Summary	Q1–Q4 comparison
Raw Data	Sample dataset

5.2 Features

- Professional formatting
 - Excel formulas (SUM, AVERAGE)
 - Currency formatting (£)
 - Embedded charts
 - Optimized column widths
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5.3 Automation

Run entire pipeline with:

```
python run_project.py
```

Final Report (Online Version):

<https://docs.google.com/spreadsheets/d/1yhF-Rxq49140DXZE5KqG4bmzuL5Ygrfq/edit>

6. HOW TO RUN

Steps

1. Open Command Prompt
2. Navigate to project folder
3. Install dependencies:

```
python -m pip install -r requirements.txt
```

4. Run:

```
python run_project.py
```

7. CONCLUSION

This project demonstrates a **complete end-to-end data analytics pipeline**, including:

- Database design (Star Schema)
 - Data cleaning and validation
 - Data analysis and visualization
 - Excel dashboard automation
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7.1 Final Results

Metric	Value
Total Revenue	£8,123,989
Transactions	125,000
Avg Order Value	£64.99
Best Product	USB Keyboard
Top Country	United Kingdom

7.2 Deliverables

- Project Documentation
 - PowerPoint Presentation
 - Excel Dashboard
 - Python Scripts
 - SQL Schema
 - Charts
 - **Final Report (Google Sheets):**
<https://docs.google.com/spreadsheets/d/1yhf-Rxq49140DXZE5KqG4bmzuL5Ygrfq/edit>
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